

Munkres 13

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1 Equivalence of Covering Spaces

Suppose B, E, E' are path connected and locally path connected. Let $p : (E, e_0) \rightarrow (B, b_0), p' : (E', e'_0) \rightarrow (B, b_0)$ be covering maps. Let $H = p_*(\pi_1(E, e_0)), H' = p'_*(\pi_1(E', e'_0))$. Then p, p' are equivalent iff $H_0 = H'_0$. Further, the equivalence is unique.

Suppose p, p' are equivalent, ie there exists a homeomorphism $h : E \rightarrow E'$ such that $p'h = p$ and $h(e_0) = e'_0$. By definition, h is the lifting of p at e'_0 with respect to p' , so $p_*(\pi_1(E, e_0)) \subset p'_*(\pi_1(E', e'_0))$. We also have $ph^{-1} = p'$, so $p'_*(\pi_1(E', e'_0)) \subset p_*(\pi_1(E, e_0))$, so $H = H'$. The map h is unique by the previous part.

Now suppose $H = H'$. Then in particular $p_*(\pi_1(E, e_0)) \subset p'_*(\pi_1(E', e'_0))$, so there exists the lifting $h : E \rightarrow E'$ such that $p'h = p$. Similarly, there exists the lifting $h' : E' \rightarrow E$ such that $ph' = p'$. It follows that $p'hh' = ph' = p'$, so hh' the lifting of p' at e'_0 wrt p' . Since $\text{id}_{E'}$ is also the lifting, $hh' = \text{id}_{E'}$. Similarly, $h'h = \text{id}_E$, so h, h' are homeomorphisms.

2 The Universal Covering Space

(a). Let X, Y, Z be locally connected and path connected spaces. Suppose $p : X \rightarrow Z, q : X \rightarrow Y, r : Y \rightarrow Z$ are continuous and p, r are covering maps. Then q is also a covering map.

It was shown that q is surjective. Let $y_0 \in Y$ and $z_0 = r(y_0)$. Pick $W(z_0)$ path connected and evenly covered by p, r . Let $\{U_\alpha\}_{\alpha \in J}, \{V_\beta\}_{\beta \in K}$ be the partitions of $p^{-1}(W), r^{-1}(W)$ into slices, respectively and in particular let $y_0 \in V_{\beta_0}$.

Observe that $q(U_\alpha) \subset r^{-1}rq(U_\alpha) = r^{-1}p(U_\alpha) = r^{-1}(W)$. Note that the slices V_β are path components of $r^{-1}(W)$ since each $V_\beta \cong W$ is open path connected. Denote by $J' = \{\alpha \in J : q(U_\alpha) \cap V_{\beta_0} \neq \emptyset\}$. Note that $q^{-1}(V_{\beta_0}) \subset \bigcup_{\alpha \in J'} U_\alpha$.

Conversely, suppose $\alpha \in J'$. Since $U_\alpha \cong W$ is path connected, by maximality of V_{β_0} , we have $q(U_\alpha) \subset V_{\beta_0}$. It follows that $q(\bigcup_{\alpha \in J'} U_\alpha) = \bigcup_{\alpha \in J'} q(U_\alpha) \subset V_{\beta_0}$, so $\bigcup_{\alpha \in J'} U_\alpha \subset q^{-1}(V_{\beta_0})$, so $q^{-1}(V_{\beta_0}) = \bigcup_{\alpha \in J'} U_\alpha$.

Consider the restrictions $p_0 : U_\alpha \rightarrow W, q_0 : U_\alpha \rightarrow V_{\beta_0}, r_0 : V_{\beta_0} \rightarrow W$. Then $p_0 = r_0 q_0 \rightarrow q_0 = r_0^{-1} p_0$, so $q_0 : U_\alpha \rightarrow V_{\beta_0}$ is a homeomorphism, since r_0, p_0 are. It follows that r maps each slice U_α homeomorphically to V_{β_0} , AWD.

Let $p : E \rightarrow B, r : Y \rightarrow B$ be covering maps. Suppose that E is simply connected. Then there exists a covering map $q : E \rightarrow Y$ such that $p = rq$.

Fix $b_0 \in B$. Pick e_0, y_0 such that $p(e_0) = b_0$ and $r(y_0) = b_0$. Since $p_*(\pi_1(E, e_0)) < r_*(\pi_1(Y, y_0))$, there exists a unique lifting $\tilde{p} : E \rightarrow Y$ such that $r\tilde{p} = p$. Since p, r are covering maps, so is $\tilde{p}$, so take $q = \tilde{p}$.

Let $p : E \rightarrow B$ be a covering map such that $p(e_0) = b_0$ and E is simply connected. Then there exists $U(b_0)$ such that the homomorphism induced by inclusion $i_* : \pi_1(U, b_0) \rightarrow \pi_1(B, b_0)$ is trivial.

Pick $U(b_0)$ such that p maps $V(e_0)$ homeomorphically to U , and let $h : U \rightarrow V$ be the homeomorphism. Let $j : V \rightarrow E$ be inclusion. Observe that $i = pjh$, so by functoriality, $i_* = p_* j_* h_*$. Since E is simply connected, $j_* : \pi_1(V, e_0) \rightarrow \pi_1(E, e_0)$ is trivial, so that i_* is also trivial.